



Lecture 2: Mathematical Models

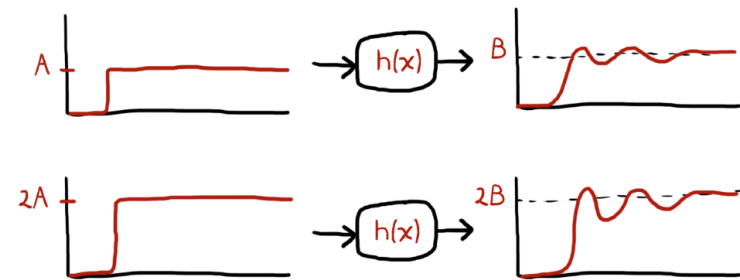
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SIST 1D#206



Linear System

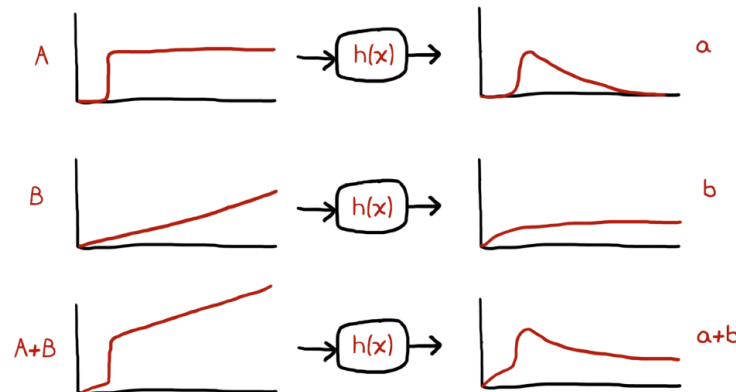
- Linearity is defined in terms of the input $u(t)$ and output $y(t)$
 - Also known as excitation and response.
- If a system satisfies **homogeneity** and **superposition**, it is a linear system.
- If a linear system further sanctifies the property of **time-invariance**, it is then called an LTI system.
 - Linearity + Time-invariance = **sinusoidal fidelity**
 - Linear system is a wider concept than *linear time-invariant (LTI)* system, also including *linear time-varying* system

Homogeneity



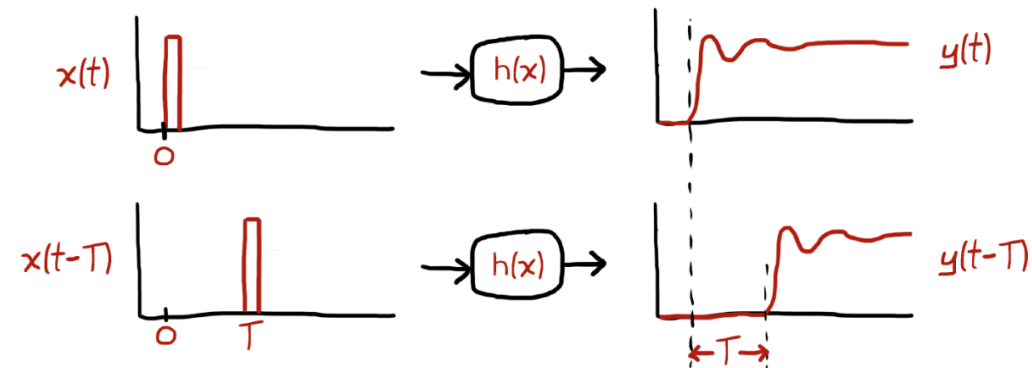
Homogeneity
 $h(ax) = a \cdot h(x)$

Superposition



Superposition
 $h(x_1) + h(x_2) = h(x_1 + x_2)$

Time Invariance



- Linear time-invariant operator

LTI Allowable operations:

Multiply or divide the input by a constant

$$a \cdot x(t) \quad \frac{1}{a} \cdot x(t)$$

Integrate or differentiate the input

$$\int x(t) dt \quad \frac{dx(t)}{dt}$$

Add or subtract multiple inputs

$$x_1(t) + x_2(t) \quad x_1(t) - x_2(t)$$

- Generally speaking, a system is also an operator (we are not ready to see this yet)

True or False:

The differential equation $\frac{d}{dt}y = -ty + u$ describes a linear system.



Laplace Transform

Excitation Functions



- Dirac delta function
- Heaviside step function

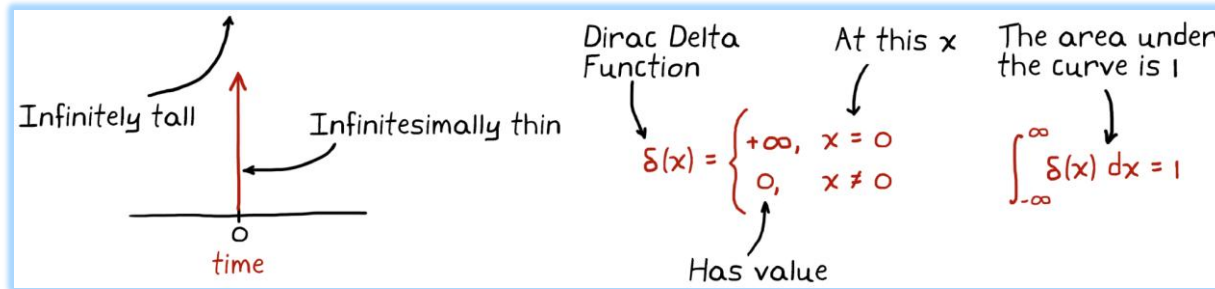


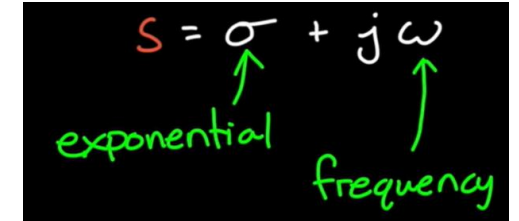
TABLE 1.1 Test waveforms used in control systems

Input	Function	Description	Sketch	Use
Impulse	$\delta(t)$	$\delta(t) = \infty$ for $0- < t < 0+$ $= 0$ elsewhere $\int_{0-}^{0+} \delta(t) dt = 1$		Transient response Modeling
Step	$u(t)$	$u(t) = 1$ for $t > 0$ $= 0$ for $t < 0$		Transient response Steady-state error
Ramp	$tu(t)$	$tu(t) = t$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Parabola	$\frac{1}{2}t^2u(t)$	$\frac{1}{2}t^2u(t) = \frac{1}{2}t^2$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Sinusoid	$\sin \omega t$			Transient response Modeling Steady-state error

- Laplace transform converts a differential equation into algebraic equation

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$

Kernel is: $K(s, t) = e^{-st}$



$s = \sigma + j\omega$
↑ exponential ↑ frequency

TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$ $u(t) = 1 \quad t > 0$ $= 0 \quad t < 0$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Example 2.1

Laplace Transform of a Time Function

PROBLEM: Find the Laplace transform of $f(t) = Ae^{-at}u(t)$.

SOLUTION: Since the time function does not contain an impulse function, we can replace the lower limit of Eq. (2.1) with 0. Hence,

$$\begin{aligned} F(s) &= \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} Ae^{-at}e^{-st} dt = A \int_0^{\infty} e^{-(s+a)t} dt \\ &= -\frac{A}{s+a} e^{-(s+a)t} \Big|_{t=0}^{\infty} = \frac{A}{s+a} \end{aligned} \quad (2.3)$$

- Laplace transform converts a differential equation into algebraic equation for its **differentiation theorem**

Example 2.2

Inverse Laplace Transform

PROBLEM: Find the inverse Laplace transform of $F_1(s) = 1/(s + 3)^2$.

SOLUTION: For this example we make use of the frequency shift theorem, Item 4 of Table 2.2, and the Laplace transform of $f(t) = tu(t)$, Item 3 of Table 2.1. If the inverse transform of $F(s) = 1/s^2$ is $tu(t)$, the inverse transform of $F(s + a) = 1/(s + a)^2$ is $e^{-at}tu(t)$. Hence, $f_1(t) = e^{-3t}tu(t)$.

TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t - T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{k-1}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

□ The two tables can be used to derive for more complicated results, e.g.,

$$F_1(s) = \frac{s^3 + 2s^2 + 6s + 7}{s^2 + s + 5}$$

$$F_1(s) = s + 1 + \frac{2}{s^2 + s + 5}$$

$$f_1(t) = \frac{d\delta(t)}{dt} + \delta(t) + \mathcal{L}^{-1}\left[\frac{2}{s^2 + s + 5}\right]$$

TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem

$$\mathcal{L}[Ae^{-at}\cos \omega t] = \frac{A(s + a)}{(s + a)^2 + \omega^2}$$

- Laplace transform converts a differential equation into algebraic equation for its **differentiation theorem**

TABLE 2.2 Laplace transform theorems

Item no.	Theorem		Name
7.	$\mathcal{L} \left[\frac{df}{dt} \right]$	$= sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L} \left[\frac{d^2f}{dt^2} \right]$	$= s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L} \left[\frac{d^nf}{dt^n} \right]$	$= s^nF(s) - \sum_{k=1}^n s^{n-k}f^{(k-1)}(0-)$	Differentiation theorem

Example 2.3

Laplace Transform Solution of a Differential Equation

PROBLEM: Given the following differential equation, solve for $y(t)$ if all initial conditions are zero. Use the Laplace transform.

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 32y = 32u(t) \quad (2.14)$$

Laplace Transform



PROBLEM: Find the Laplace transform of $f(t) = te^{-5t}$.

ANSWER:



TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$ $u(t) = 1 \quad t > 0$ $= 0 \quad t < 0$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

PROBLEM: Find the inverse Laplace transform of $F(s) = 10/[s(s+2)(s+3)^2]$.

ANSWER:



TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t-T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{(k-1)}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²



Transfer Function

- General case of a high order differential equation:

$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t) \quad (2.50)$$

- **Convert it into s-domain:**

$$\begin{aligned} a_n s^n C(s) + a_{n-1} s^{n-1} C(s) + \cdots + a_0 C(s) + \text{initial condition} \\ \text{terms involving } c(t) \\ = b_m s^m R(s) + b_{m-1} s^{m-1} R(s) + \cdots + b_0 R(s) + \text{initial condition} \\ \text{terms involving } r(t) \end{aligned}$$

- Assume zero initial conditions we get transfer function

$$\frac{C(s)}{R(s)} = G(s) = \frac{(b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0)}{(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0)}$$

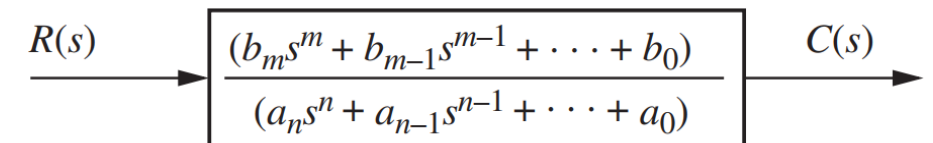


FIGURE 2.2 Block diagram of a transfer function

PROBLEM: Find the transfer function, $G(s) = C(s)/R(s)$, corresponding to the differential equation $\frac{d^3 c}{dt^3} + 3\frac{d^2 c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2 r}{dt^2} + 4\frac{dr}{dt} + 3r$.

ANSWER:

PROBLEM: Find the differential equation corresponding to the transfer function,

$$G(s) = \frac{2s + 1}{s^2 + 6s + 2}$$

ANSWER:

PROBLEM: Find the ramp response for a system whose transfer function is

$$G(s) = \frac{s}{(s + 4)(s + 8)}$$

ANSWER:



System Examples

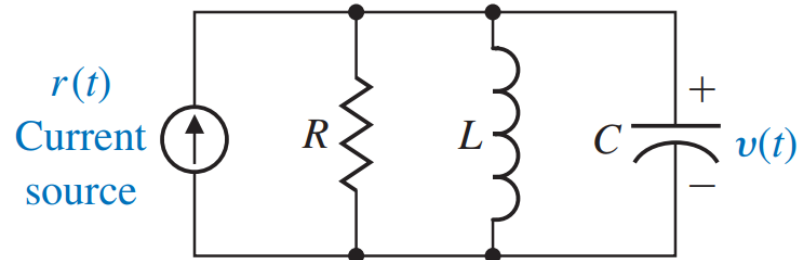
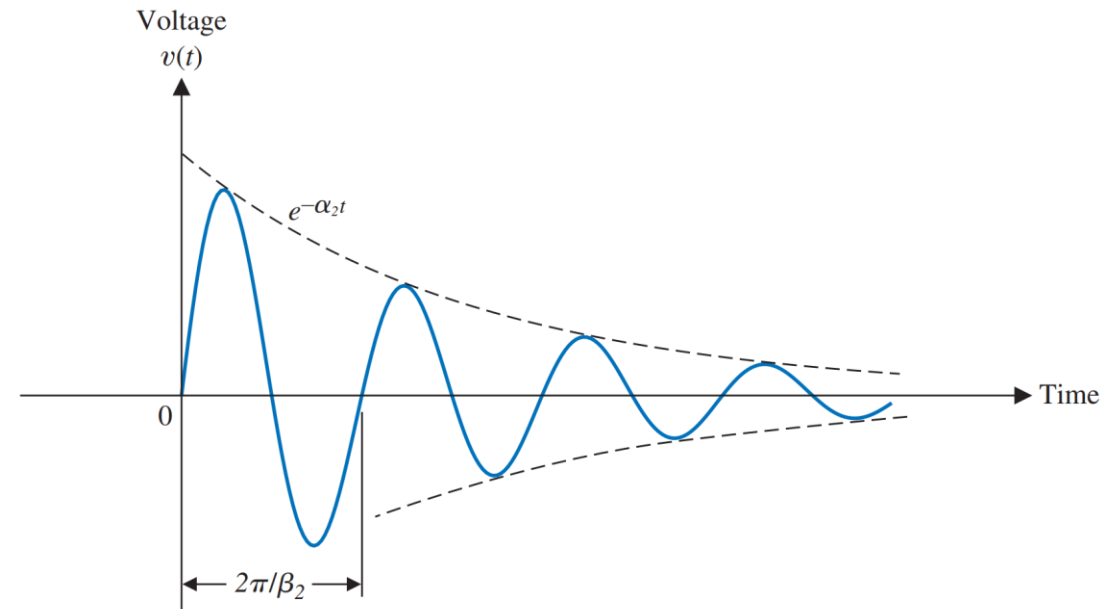


FIGURE 2.3
RLC circuit.

$$\frac{v(t)}{R} + C \frac{dv(t)}{dt} + \frac{1}{L} \int_0^t v(t) dt = r(t)$$

$$v(t) = K_2 e^{-\alpha_2 t} \cos(\beta_2 t + \theta_2)$$

FIGURE 2.4
Typical voltage
response for
an *RLC* circuit.



Spring mass damper

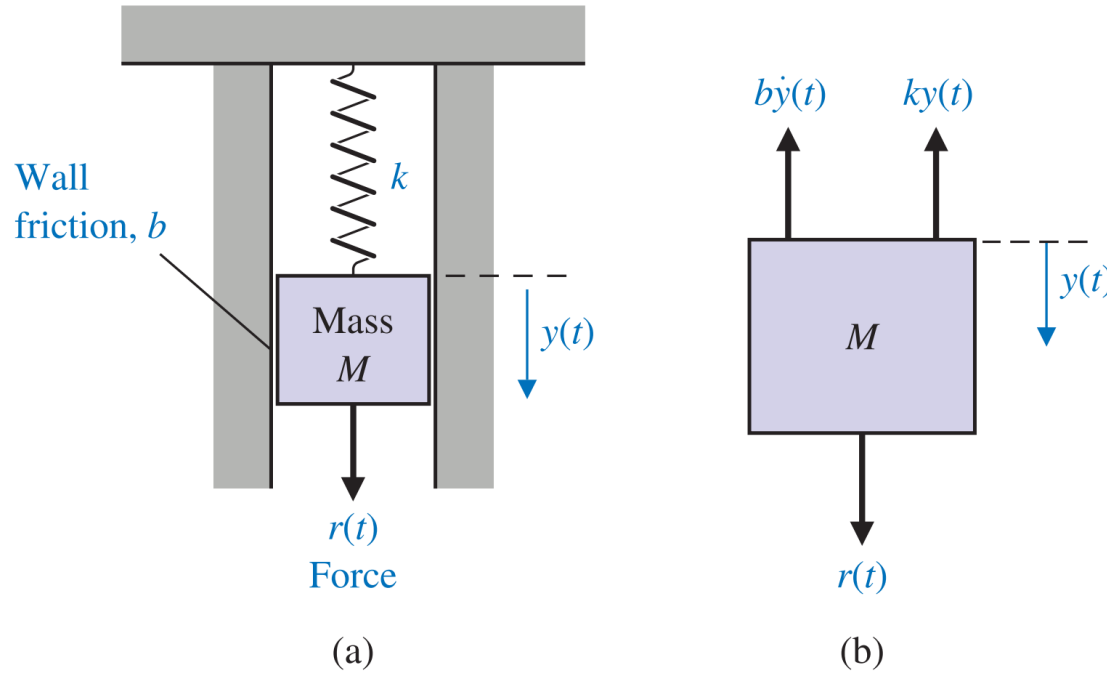


FIGURE 2.2
(a) Spring-mass-damper system.
(b) Free-body diagram.

$$M \frac{d^2 y(t)}{dt^2} + b \frac{dy(t)}{dt} + ky(t) = r(t)$$

$$y(t) = K_1 e^{-\alpha_1 t} \sin(\beta_1 t + \theta_1)$$

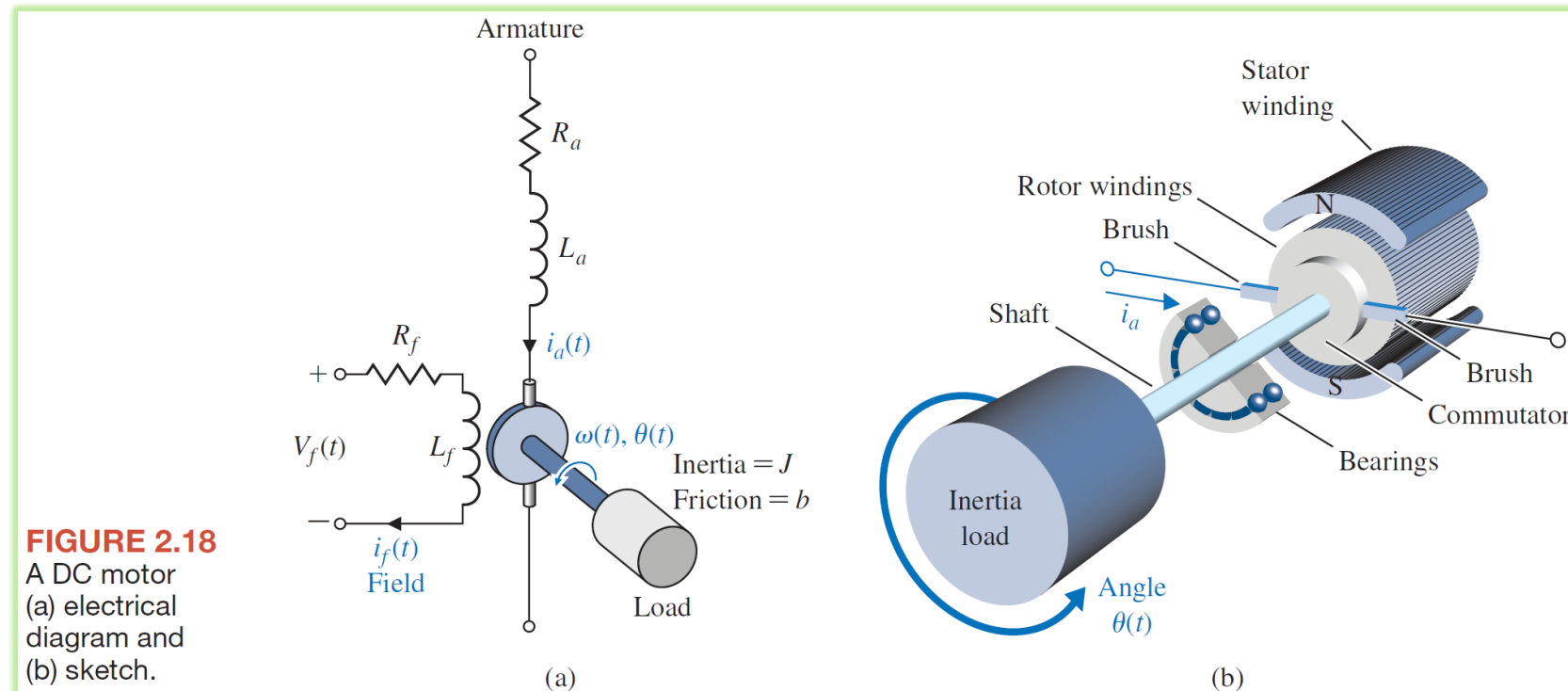
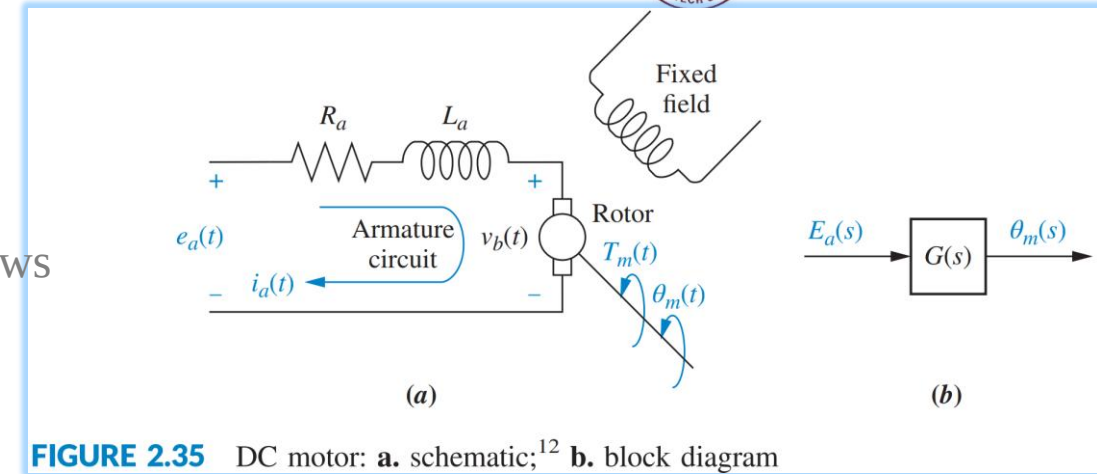
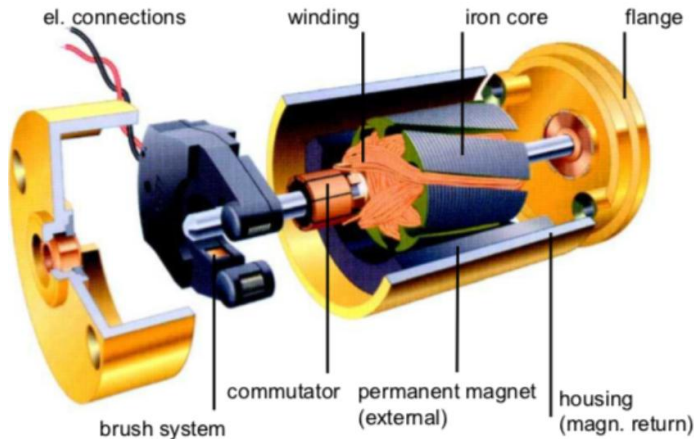
Electro-mechanical system



- ❑ Newtonian mechanics
- ❑ Faraday's law of induction
 - With a little bit of math of total differential and derivative laws
- ❑ Ampere force

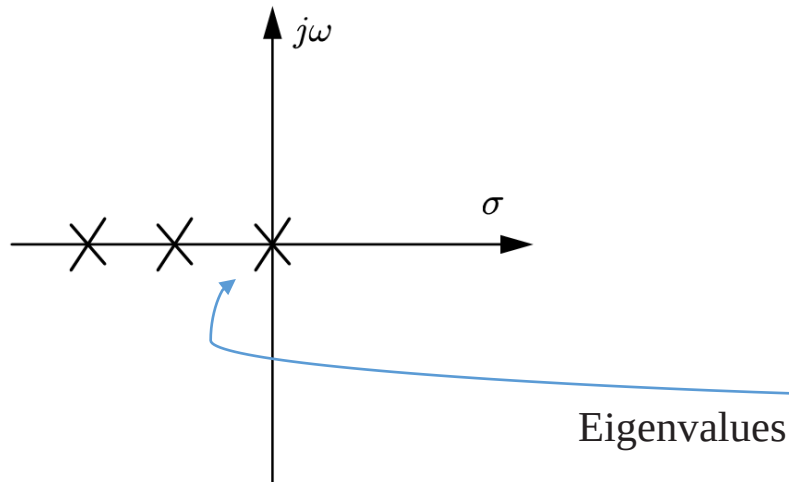
$$J \frac{d^2}{dt^2} \Theta = D \frac{d}{dt} \Theta + K_T i_1$$

$$u_1 = R i_1 + K_E \frac{d}{dt} \Theta + L_1 \frac{d}{dt} i_1$$



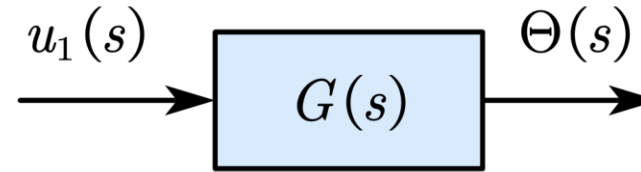
Overview of the Math Models

- ❑ O.D.E (from physics laws)
- ❑ Block diagram
- ❑ Transfer function (via Laplace transform)
- ❑ State space model
- ❑ Pole zero map (related to eigenvalues)



$$J \frac{d^2}{dt^2} \Theta = D \frac{d}{dt} \Theta + K_T i_1$$

$$u_1 = R i_1 + K_E \frac{d}{dt} \Theta + L_1 \frac{d}{dt} i_1$$



Can you give details within $G(s)$?

$$\frac{d}{dt} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & K_T J^{-1} \\ 0 & -K_E L^{-1} & -R L^{-1} \end{bmatrix} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ L^{-1} \end{bmatrix} u + \begin{bmatrix} 0 \\ J^{-1} \\ 0 \end{bmatrix} \tau_L$$



Exercise for modeling of an e.m. system



Example 2.23

Transfer Function—DC Motor and Load

PROBLEM: Given the system and torque-speed curve of Figure 2.39(a) and (b), find the transfer function, $\theta_L(s)/E_a(s)$.

SOLUTION: Begin by finding the mechanical constants, J_m and D_m , in Eq. (2.153). From Eq. (2.155), the total inertia at the armature of the motor is

$$J_m = J_a + J_L \left(\frac{N_1}{N_2} \right)^2 = 5 + 700 \left(\frac{1}{10} \right)^2 = 12 \quad (2.164)$$

and the total damping at the armature of the motor is

$$D_m = D_a + D_L \left(\frac{N_1}{N_2} \right)^2 = 2 + 800 \left(\frac{1}{10} \right)^2 = 10 \quad (2.165)$$

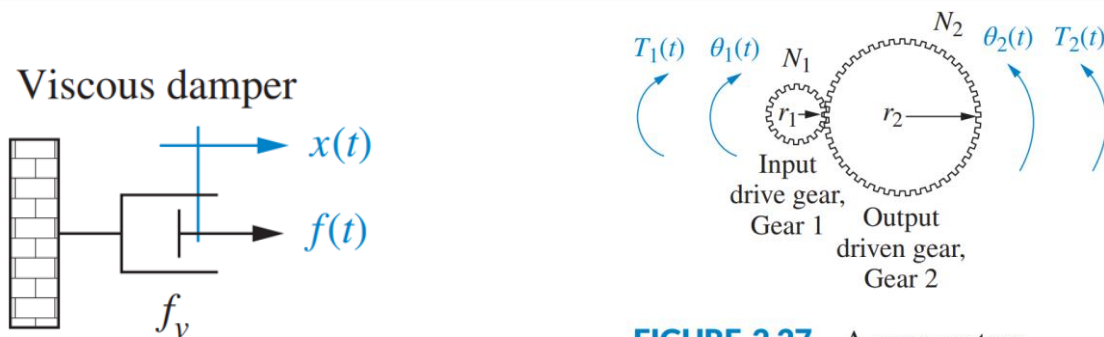


FIGURE 2.27 A gear system

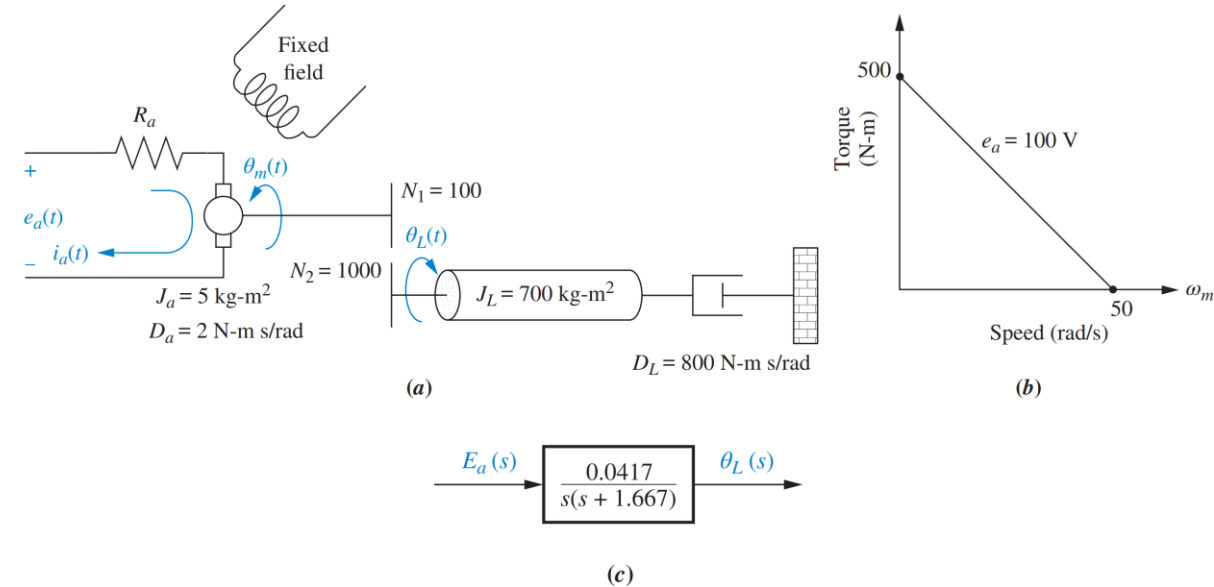


FIGURE 2.39 a. DC motor and load; b. torque-speed curve; c. block diagram

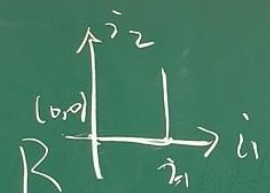
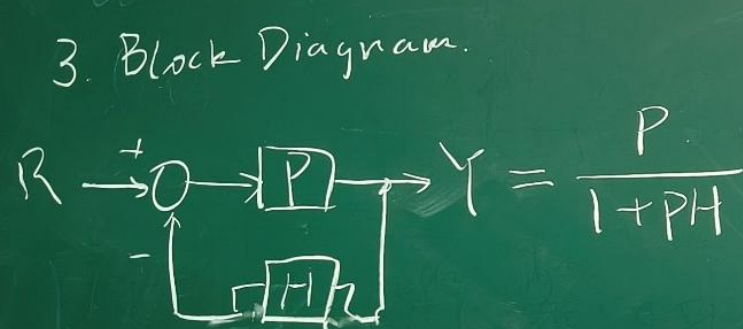
Generalizing the results, we can make the following statement: *Rotational mechanical impedances can be reflected through gear trains by multiplying the mechanical impedance by the ratio*

$$\left(\frac{\text{Number of teeth of gear on destination shaft}}{\text{Number of teeth of gear on source shaft}} \right)^2$$

where the impedance to be reflected is attached to the source shaft and is being reflected to the destination shaft. The next example demonstrates the application of the concept of reflected impedances as we find the transfer function of a rotational mechanical system with gears.

System Model

1. O.D.E $\rightarrow$ initial conditions set to zero $\left(\frac{Y(s)}{R(s)} = \frac{S^2}{s+1} \right)$ (not proper)
2. Transfer Function (only describe a pair of I/O relation)
3. Block Diagram.



Force-Voltage Analogy



$$U - Ri = L \frac{di}{dt} \rightarrow \text{Faraday's Law}$$

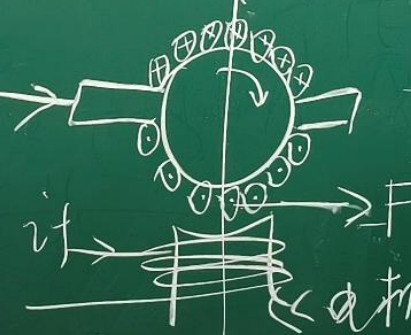
$$T = J \frac{d\omega}{dt} = K_m \cdot i_f(t) - b\omega$$

$$T_m \dot{p} = m \dot{v}$$

4. Signal Flow Graph

$$d\psi_1(i_1, i_2) = \frac{\partial \psi_1}{\partial i_1} di_1 + \frac{\partial \psi_1}{\partial i_2} di_2$$

$$\psi_1 = L_1 i_1 + L_2 i_2$$



$$e = \frac{d\psi}{dt} = \frac{d(Li)}{dt}$$

$$= \frac{dL}{dt} i + L \frac{di}{dt}$$

$$= \frac{\partial L}{\partial \phi} \omega \cdot i + L \frac{di}{dt}$$